◆ RESEARCH PROPOSAL

Research Proposal Template — PhD Scholarships

A structured template for writing research proposals for MEXT, KGSP, RTP, and DAAD PhD scholarships — with annotated examples.

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Resource #09 of 09

Inside This Template

1. What This Template Covers (and Doesn't)

This template is designed for **PhD-level research proposals submitted as part of fully-funded scholarship applications** — primarily MEXT (Japan), DAAD Research Grants (Germany), CSC (China), KGSP/GKS (Korea), and RTP (Australia).

Length & Format Targets

Scholarship	Required Length	Style
MEXT (Field of Study & Study Plan)	2 pages	Concise, structured, study-plan-heavy
DAAD Research Grant	5–10 pages	Academic, methodology-heavy, references
CSC (Study Plan)	3–5 pages	Structured, bilingual-friendly, year-by-year plan
GKS / KGSP (Study Plan)	3–5 pages	Academic + cultural fit narrative
Australia RTP / Direct PhD Apps	5–8 pages	Academic, highly methodological

What this template does NOT replace: a Statement of Purpose (SOP) or Motivation Letter. Those documents are *about you*; the research proposal is *about the research*. Use ScholarMap's separate SOP Template Pack alongside this one.

Universal Structure

Every academic research proposal follows roughly the same 8-section skeleton. Length per section shifts based on the scholarship's word limit, but the order rarely does:

1. Title + Abstract → **2. Background** → **3. Research Question** → **4. Framework** → **5. Methodology** → **6. Timeline** → **7. Expected Outcomes** → **8. References**.

2. How Reviewers Actually Score Your Proposal

Most scholarship rubrics are not public. But based on rubrics that have been published (DAAD, NSF, Wellcome, ARC) and conversations with internal reviewers, evaluation breaks into 5 components:

Criterion	Approx. Weight	What Reviewers Check
Originality of research question	20–25%	Does this question already have an answer? Is the gap real?
Methodological rigour	25–30%	Are the methods justified? Realistic? Field-standard?
Feasibility	15–20%	Can this actually be done in the funding period?
Significance / contribution	15–20%	Why does anyone outside academia care?
Writing quality + structure	10–15%	Can a non-specialist follow the argument?

The 'Three-Person Test'

Before submitting, your proposal should pass three different readers:

1. **The specialist:** Can a researcher in your subfield identify the gap? Are your methods rigorous?
2. **The generalist:** Can a PhD in a neighbouring field follow the argument?
3. **The administrator:** Can the embassy reviewer (non-PhD) understand significance + feasibility?

Most failed proposals fail Test 2 — too jargony for the generalist on the panel. Always have at least one non-specialist read your proposal before submission.

3. Section 1 — Title & Abstract

Title (12–18 Words)

Strong proposal titles do three things in one sentence: name the **phenomenon**, the **method**, and the **context**.

Examples — Weak vs Strong

Weak	Strong
A Study of Renewable Energy	Quantifying the Productivity Spillover of Solar Microgrids in Off-Grid Sub-Saharan Comm
Composite Materials Research	Humidity-Induced Degradation of Fibre-Reinforced Polymer Blades in Equatorial Climat
Machine Learning for Healthcare	A Few-Shot Diagnostic NLP Pipeline for Low-Resource African Languages: Evaluation

Abstract (150–250 Words)

The abstract is what reviewers read first — and often the only thing they read in detail before scoring. Use this 6-sentence skeleton:

Sentence 1: The big-picture context (1–2 lines).
Sentence 2: The specific gap or unanswered question.
Sentence 3: Your research question (state it explicitly).
Sentence 4: Your methodology in one breath (data + method + sample).
Sentence 5: Expected output / contribution.
Sentence 6: Why it matters beyond academia.

Sample Abstract

Renewable energy expansion in Sub-Saharan Africa has prioritised rural electrification, yet the productivity gains promised to communities remain weakly evidenced beyond pilot studies. Existing impact assessments rely on aggregate national data and miss intra-village heterogeneity. ***This study asks how solar microgrid uptake reshapes household-level productivity over 5–7 years*** *in three off-grid Nigerian states. Using a difference-in-differences design with monthly satellite luminosity data and 1,200 panel-survey households (2018–2025), I will identify causal productivity spillovers conditional on microgrid scale and tariff structure. Findings will inform Nigeria's National Electrification Project and similar initiatives across ECOWAS, providing the first peer-reviewed evidence base for tariff-design decisions affecting 70 million off-grid Africans.*

4. Section 2 — Background & Literature Review

This section establishes **why your question is worth asking**. It is not a literature dump — it is a focused argument that ends with the gap your proposal fills.

Structure (1.5–2 Pages)

Para 1 — Big context (5–7 lines): The broad field, current state, and why anyone is paying attention.
Para 2 — Sub-field state of the art (8–12 lines): What has been done, by whom, with what methods. Cite 8–15 sources.
Para 3 — Active debates (5–7 lines): Where do scholars disagree? Cite 2–3 competing positions.
Para 4 — The gap (5–7 lines): What has NOT been studied? Why does it matter? End with one sentence: 'This proposal addresses [gap] by [approach].'

Citation Style by Field

Field	Default Citation Style	Reference Manager
Sciences / Engineering / Medicine	IEEE, Vancouver, AMA, ACS	Mendeley, Zotero, EndNote
Social Sciences / Humanities	APA, Chicago, MLA, Harvard	Zotero, Mendeley
Economics / Business	APA, Harvard	Zotero, EndNote
Law	OSCOLA, Bluebook	Zotero, EndNote

Pro tip: Use a citation manager from Day 1. Manual reference formatting is the #1 time sink in proposal writing — and reviewers spot inconsistent styles instantly.

How Many Sources to Cite

For a 5–10 page proposal, expect **20–40 cited references**. Distribution by recency:

- **50%** from the last 5 years (shows you are current).
- **30%** from the last 5–10 years (foundational recent work).
- **20%** seminal / older works (the field-defining classics).
- At least 1–2 sources from the lab/researchers at your target host university (signals fit).

5. Section 3 — Research Question(s) & Hypotheses

This section is short (1/2 page maximum) but high-stakes. A weak research question kills strong proposals.

The 1 + 3 Structure

Most successful proposals use one **main research question** followed by 2–3 **sub-questions**:

RQ (Main): [One clear, answerable question — usually 'How does X affect Y under condition Z?']
SQ1: [What is the magnitude of the effect?]
SQ2: [What mechanism explains the effect?]
SQ3: [Under what conditions does the effect change?]

Question Quality Test (FINER)

Use the FINER framework — every research question must be:

- **F — Feasible:** Can be answered with available time, data, methods.
- **I — Interesting:** Other researchers care about the answer.
- **N — Novel:** Not already definitively answered.
- **E — Ethical:** Can be done without causing harm.
- **R — Relevant:** Generates knowledge that advances theory or practice.

Hypotheses (Quantitative Proposals Only)

If you are doing quantitative work, state hypotheses explicitly. Each main RQ becomes one or two hypotheses:

H1: Solar microgrid access increases monthly household productive output by ≥15% within 2 years post-installation.
H2: The productivity effect is conditional on tariff structure — flat-rate tariffs produce significantly larger gains than per-kWh tariffs.
H3: The productivity effect attenuates over time as initial enterprise formation saturates local markets.

Qualitative proposals replace hypotheses with **guiding propositions** — open statements that the data can refine but not formally test.

6. Section 4 — Theoretical / Conceptual Framework

This section (about 1 page) names the **theoretical lens** through which you will interpret your data. Reviewers want to see you are working *within* a tradition, not free-floating.

Three Common Frameworks by Field

Field	Common Frameworks	Example Application
Economics	Solow Growth Model, Game Theory, Behavioural Economics	Productivity spillover via Solow + behavioural adoption barriers
Sociology	Social Capital Theory, Bourdieu's Capitals, Network Theory	Migration network effects on labour-market entry
Engineering	Failure Mode Analysis, Systems Theory, Reliability Theory	FMEA for tropical-climate FRP degradation
Public Health	Social Determinants of Health, Health Belief Model	Vaccine uptake via Health Belief Model in rural Tanzania
Computer Science	Information Theory, Bayesian Inference, Optimisation Theory	Few-shot NLP via Bayesian probabilistic framing

How to Write This Section

Para 1: Introduce the framework + the 1–2 seminal authors who developed it.
Para 2: Explain how it has been applied in your field (cite 2–3 papers).
Para 3: Justify why it is appropriate for YOUR research question.
Para 4: Operationalise — what variables, constructs, or relationships does the framework specify that you will measure?

Common mistake: dropping a framework name (e.g., 'I will use Bourdieu') without explaining how. Reviewers want to see the framework's *operational implications*: which variables come from where, and how they connect.

7. Section 5 — Methodology

The single most important section. Reviewers will spend 60–70% of their evaluation time here. Allocate 1.5–2 pages.

Standard Subsections

Subsection	Length	Content
5.1 Research Design	1/3 page	Quantitative? Qualitative? Mixed methods? Justify.
5.2 Setting / Population	1/4 page	Where, who, why this sample/site
5.3 Data Collection	1/2 page	Sources, instruments, procedures, sample size
5.4 Analytical Strategy	1/2 page	Specific techniques, software, statistical/analytical approach
5.5 Validity & Reliability	1/4 page	How you address bias, error, generalisability
5.6 Ethical Considerations	1/4 page	IRB approval, consent procedures, data protection

5.3 Data Collection — Be Specific

Reviewers want concrete answers, not vague intentions:

✗ **Vague:** 'I will collect data through surveys.'
✓ **Specific:** 'Primary data: 1,200 face-to-face structured surveys (60-minute, Yoruba/English) across 12 villages, conducted June–August 2026 by 6 trained enumerators using Open Data Kit on tablets, with audio backup. Secondary data: monthly satellite luminosity 2018–2025 from VIIRS DNB and quarterly tariff records from the Rural Electrification Agency.'

5.4 Analytical Strategy — Show Method Selection

Reviewers reward applicants who justify *why* a method, not just *which* method:

> *'I will use a difference-in-differences (DiD) estimator with two-way fixed effects, comparing treated villages (post-microgrid) with matched controls. DiD is preferred over RCT given ethical concerns with random allocation of public infrastructure, and over OLS given likely selection bias in microgrid placement. I will run robustness checks using synthetic control method (Abadie 2010) and event-study designs.'*

8. Section 6 — Timeline & Feasibility

Use a **quarter-by-quarter or semester-by-semester table**. Reviewers use this section to check whether your plan is *realistic* for the funding period. Generous estimates fail; pessimistic ones do too.

3-Year PhD Timeline Template

Quarter	Months	Activities	Milestones / Outputs
Q1	1–3	Coursework + literature review	Confirmation of research design
Q2	4–6	Pilot study + ethics approval + instrument design	Ethics clearance + pilot report
Q3	7–9	Pre-registration + first round of data collection	Conference abstract submission
Q4	10–12	Initial data analysis + first paper draft	Annual review pass
Q5	13–15	Second wave of data collection / experiments	Working paper online
Q6	16–18	Mid-thesis chapter writing	First paper journal submission
Q7	19–21	Third wave / robustness checks	Conference presentation
Q8	22–24	Mid-thesis review	Mid-thesis defence pass
Q9	25–27	Chapter 5–6 writing	Second paper draft
Q10	28–30	Chapter 5–6 revision + Chapter 7 writing	Second paper submission
Q11	31–33	Full thesis revision	Internal review pass
Q12	34–36	Final defence prep + thesis submission	PhD defended + degree awarded

Feasibility Statement

Add 2–3 sentences explicitly stating why the timeline is feasible: (1) data already accessible, (2) pilot study already done, (3) co-supervisor relationship established, (4) tools/software available, (5) IRB precedent in field.

9. Section 7 — Expected Outcomes & Contribution

This section (1/2 page) answers: **'When this PhD is finished, what new things exist in the world?'**

The 4-Layer Contribution

Layer	Question	Example
Theoretical	What new knowledge?	First quantitative test of solar-microgrid productivity spillovers in West Africa
Methodological	What new way of measuring?	Open-source DiD estimator pipeline for satellite + survey data
Empirical	What new evidence?	Panel dataset of 1,200 households, 7 years, publicly archived
Practical / Policy	Who uses this beyond academia?	Tariff-design recommendation memo to Nigeria's REA

Expected Publications & Outputs

Most strong proposals commit to:

- 2–3 peer-reviewed journal articles (named target journals).
- 1–2 international conference presentations (named conferences).
- 1 policy brief or industry white paper.
- 1 open dataset / open-source code release (where ethical).
- Public-facing output: blog post, op-ed, or short documentary.

Naming target journals signals you understand the field's publication landscape. For example: 'Targeting submission to Energy Economics, World Development, and Energy Policy.'

10. Section 8 — References & Annexes

References

• Use one consistent citation style — chosen by your field, not by you.
• Generate via citation manager (Zotero / Mendeley / EndNote) — never type manually.
• Include 20–40 references for a 5–10 page proposal.
• Each reference cited in-text must appear in the list, and vice versa.
• Sort alphabetically by first author surname, OR numerically by appearance — match your field's convention.

Useful Annexes (Optional)

Some scholarships allow appendices. Use them sparingly:

- **Annex A:** Detailed methodology (full survey instrument, equation derivations).
- **Annex B:** Full timeline Gantt chart.
- **Annex C:** CV of supervisor / co-supervisor.
- **Annex D:** Letter of access from data-providing institution.
- **Annex E:** Pilot study results (1–2 pages).

Annexes do NOT replace weak main text. If reviewers must read your annex to understand your methodology, your main methodology section is too weak.

11. Annotated Sample Proposal Outline

Below is the full outline of a successful DAAD Research Grant PhD proposal (Renewable Energy Economics, awarded 2024). Page numbers and section weights illustrate balanced structure.

Section	Pages	Word Count	Key Move
Title + Abstract	0.5	210	Names region, period, method in 16 words
1. Background & Lit Review	1.7	850	8 sources from last 5 years; 4 from supervisor's own group; 3 competi
2. Research Questions	0.4	180	1 main + 3 sub-questions; FINER-tested
3. Conceptual Framework	0.8	400	Solow growth + behavioural adoption — operationalised into 6 measu
4. Methodology	1.9	950	DiD design + 1,200 panel households + ODK survey + STATA pipelin
5. Timeline	0.6	280 + table	12 quarters, 12 milestones, 2 contingency notes
6. Expected Outcomes	0.5	260	3 papers (Energy Economics, World Development, Energy Policy) + 1
7. References	0.6	—	32 references, IEEE style, Mendeley-generated
Total	7.0	~3,130	

Why This Proposal Won

✓ Balanced section weights — methodology was the longest section, as DAAD reviewers expect.
✓ Cited 4 papers from supervisor's own group + 8 recent papers from the broader field.
✓ Pilot study (referenced as Annex E) gave evidence of feasibility.
✓ Named specific target journals — signalled publication strategy.
✓ Practical contribution (REA policy brief) extended impact beyond academia — DAAD scoring criterion.

Final Word

A PhD research proposal is not a description — it is an argument. The argument has 5 moves: (1) the field has a question, (2) here is what we don't know, (3) here is what I will ask, (4) here is how I will answer it, (5) here is why the answer matters.

If your reader can summarise those 5 moves in one breath after reading your proposal — you've won the document round.

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